

Well, of course, I can't guarantee that. Remember, this is purely a hypothetical example in which the correlation coefficient is artificially set to -100% . In the real world, you can never really find such a situation. Think again about the example. If one investment does better than average by $x\%$, when the other performs worse than average by $x\%$, then the good return more or less nullifies the bad return, and you are left with the average. In sum, the perfect negative correlation means that any bad surprises from one investment will be offset by good surprises from the other asset. Put them both together in the same portfolio, and you have no surprises.

Now, you may be thinking that if the two investments are perfectly correlated in a negative fashion, why don't they completely cancel each other out, leaving you nothing at the end of the year? If one completely zigs, when the other completely zags, shouldn't you be left with a flat profile?

Not exactly. Remember that negative or positive correlations are rarely 100% . More importantly, I did not say that when one investment goes up 8% , the other goes down 8% . I said that when one goes up by more than the average performance of the market, the other goes down by more than the average performance of the market.

In Table 3.1, I assumed both investments are expected to increase by 15% per year. The perfect negative correlation therefore means that when one market goes up 23% (8% more than the 15% average expected), the other will increase by only 7% (8% less than the 15% average expected). In other words, one has climbed 8% more than average, while the other has risen 8% less than average. But both have gone up.

Enough with perfect correlations. Let's take a look at the more reasonable and more common middle ground. In other words, let's see what happens when you diversify into investments that have correlations that are much greater than -100% and much smaller than $+100\%$. For example, in Table 3.1, you will see that if the correlation coefficient is exactly zero—that is, there is absolutely no relationship between the two investments' movements—the probability of earning less than a 5% CD is roughly 28% . Compare that with the 37% chance of doing worse than the CD by having all your eggs in one basket.